Assignment 1

Robotics in Medicine and Industry

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Contents

1 Executive Summary **2**

2 Task 1: Literature Survey **3**
2.1 Conventional Bone Cutting Tools . . . 3
2.2 Advanced Bone Cutting Technologies . . . 5

3 Task 2: Conformal Tool Design **6**
3.1 Dual-Axis Articulating Burr . . . 6
3.1.1 Dual Axis . . . 6
3.1.2 Extender . . . 6
3.2 Surgical Workflow . . . 6
3.3 CAD Design . . . 7
3.3.1 Overview of the End-Effector Design . . . 8
3.3.2 Operation Workflow of the End-Effector . . . 10
3.4 Safety Precautions . . . 12
3.4.1 Ultrasonic Sensor . . . 12

4 Task 3: Surgical Planning **14**
4.1 Parameterize the Cuts . . . 14
4.1.1 Position (Entry Point Calculation) . . . 14
4.1.2 Principal Component Analysis (PCA) for Tumor Orientation . . . 14
4.1.3 Determining the Radius at the Cut Layer . . . 15
4.1.4 Adding a Safe Offset . . . 15
4.1.5 Calculating the Entry Point . . . 15
4.1.6 Determining the Final Cut Point . . . 15
4.2 The Orientation of the Tool . . . 16
4.2.1 Tilt Angle (θ) . . . 16
4.2.2 Rotation Angle (ϕ) . . . 16
4.3 Determine the Cutting Depth at Each Layer . . . 16
4.4 Circular Path Planning . . . 17
4.5 Rectangular Blocks for Tumor Removal . . . 17
4.6 Position: Entry and Exit Points . . . 17
4.7 Orientation: Tool Alignment for Cutting . . . 17
4.8 Width of the Block . . . 18
4.9 Length of the Block . . . 18
4.10 Breadth of the Block . . . 18

5 Pseudocode for Parametrization of the Cutting Path **18**

6 Algorithm Implementation on Given Dataset **20**

7 Conclusion **21**

1 Executive Summary

Bone tumors present a significant medical challenge, affecting many lives due to their historical treatment methods, anatomical location, and complex diagnosis. These tumors are classified as either benign (non-cancerous) or malignant (cancerous). Historically, malignant bone tumors were treated through amputation, effectively reducing the risk of secondary bone cancer. However, these procedures often resulted in severe physical limitations and psychological distress for patients.

The management of bone tumors has been revolutionized by conformal bone surgery, also known as limb-sparing surgery. This technique removes malignant tumors while preserving the patient's physical function, addressing key postoperative concerns associated with amputation. Advances in technology have enabled the development of patient-specific, 3D-printed implants for precise bone reconstruction. Preoperative imaging techniques such as CT scans and MRIs provide high-resolution images of the affected bone and surrounding skeletal structure. This data is then processed using CAD to model customized implants.

To ensure the implant accurately matches the resected bone, precise and controlled cuts must be made in the operating room (OR). Selecting appropriate tools for the specific bone type and determining the necessary cutting parameters are crucial for successful bone replacement, ultimately allowing patients to regain normal function.

This report aims to provide information on the different tools required for such procedures, propose a tool design suitable for specific bone tumor shapes, and outline a surgical plan for its application.

2 Task 1: Literature Survey

2.1 Conventional Bone Cutting Tools

Conventional bone-cutting tools are widely used in surgical procedures due to their effectiveness, ease of use, and cost-efficiency. Among the most common tools are saws, drills, burrs, and milling devices. Saws—including reciprocating, oscillating, and wire saws—are typically coated with diamond or carbide to enhance wear resistance, lifespan, and rigidity. They are frequently used in orthopaedic surgeries such as knee and hip arthroplasties, allowing surgeons to make controlled bone cuts. Since their periodic motion is linear and features small teeth along the periphery, surgeons can precisely control both the cutting depth and direction. These saws are usually made of stainless steel, high-speed steel, or high-carbon steel. Additionally, a saw removes a slit of material approximately equal to the thickness of the blade [1], resulting in a small volume of bone being removed. However, this process can contribute to surgeon fatigue during lengthy procedures. Bone has a low thermal conductivity of $0.4W/mK$, meaning heat accumulates rather than dissipates. Consequently, a major limitation of sawing is excessive heat generation, which can lead to *thermal necrosis*—the death of bone cells due to high temperatures—requiring precise manual handling to avoid damaging adjacent tissues. Due to the continuous contact between the saw blade and the bone surface, heat accumulates, potentially raising surface temperatures above 150°C, which can prolong the healing process [2]. Moreover, sawing generates bone debris, commonly referred to as 'chips,' which may obscure the surgeon's field of visibility. Regarding the precision involved in sawing, the continuity of mass must be considered, as described by Eq. 1. This equation defines the relationship between the sawing speed V_s, the thickness of the uncut workpiece (bone) layer h_o, the chip velocity V_c, and the chip thickness h_c [2].

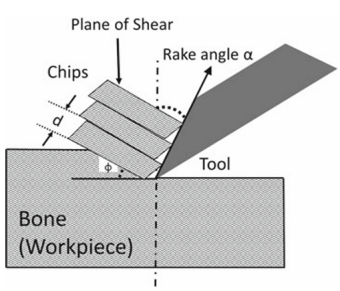


Figure 1: The left image shows a schematic of the mechanism of saw cutting [2], while the right image is an X-ray of a bone showing signs of thermal necrosis, with a highlighted region indicating affected bone tissue[1].

$$V_s h_o = V_c h_c [2] \tag{1}$$

The precision of sawing is diminished by occurring vibrations, disrupting the uniformity of the saw speed and chip thickness. The fluctuations of V_s and h_o causes inconsistency in the chip thickness (h_c) leading to bone microfractures, an irregular bone surface and higher cutting forces, ultimately, affecting the stability of the implant and the lifespan of the tool. Therefore, to prevent the temperature of the tool and the bone from rising above the threshold for osteonecrosis surgeons must reduce the frequency (speed) of their cuts.

$$\Gamma = \frac{V_{shear}}{d} [2] \tag{2}$$

High shear strain, Γ, would increase the force transmitted to the bone, thus making the bone more susceptible to microfractures, additionally, a high shearing velocity, V_{shear}, would increase the thermal damage on the bone.

Drills and burrs are essential tools in bone surgery, particularly for creating holes for implants and shaping bone structures. The most common type is the standard-point twist drill, which features helical grooves [2]. Drilling is generally a fast and efficient method, allowing for consistent hole diameters and depths—critical for implant stability. The speed and force of the drill can be adjusted based on bone density. However, drilling generates significant heat due to friction and bone deformation during cutting. Bone is an anisotropic material, meaning its strength varies with orientation. The concentrated force applied during drilling can lead to microcracks, increasing the risk of osteonecrosis (bone tissue death), ultimately weakening the bone [3]. To prevent unintended bone tissue damage, it is necessary to optimise drilling parameters, such as drill diameter, rotational speed, and axial forces, according to the specific bone characteristics. Additionally, cooling techniques, including external cooling of the drilling area and internal coolant lubrication within the drill's grooves, as depicted in Fig. 2, help maintain a safe bone temperature and minimise thermal damage [3].

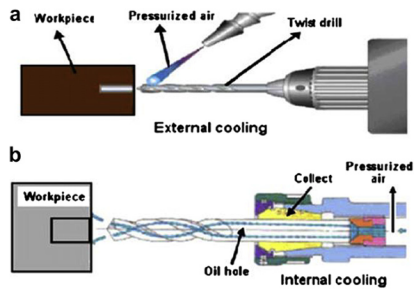


Figure 2: (a) External cooling and (b) Internal cooling

Milling tools, which are often used in conjunction with robotic systems, provide a means of achieving highly precise bone shaping, particularly in joint replacement procedures. They are a versatile method of material removal since they can perform cuts with a variety of paths and geometries [1], as seen in Fig. 3:

(a) Types of mills [2]

(b) Schematic of milling [2]

Figure 3: Schematic of mills

The removal of bone occurs whenever a milling tooth is forced onto the bone. This process happens gradually, making a mill less efficient than a drill due to its lower material removal rate. However, the incremental removal of material reduces heat buildup on the bone, minimising the risk of osteonecrosis. Additionally, mills have more moving parts and multiple axes of motion, making the process harder to control. On the other hand, this multi-directional cutting capability and variety in geometry allow for greater contouring and customisation in bone shaping. However, milling requires careful planning and is less commonly used as a standalone tool in manual surgeries. An important drawback of saws, drills, and burrs is the significant bone loss they cause due to their cutting diameter, which is typically at least 1.5–2 mm [4]. While these conventional tools remain essential in surgical practice, their limitations have driven the development of more advanced cutting technologies.

Figure 4: Various conventional bone cutting tools used in orthopedic surgeries, arranged from left to right: (1) Oscillating Saw - Used for controlled bone cutting. (2) Burr - Used for precise bone shaping and contouring. (3) Milling Device - Used for bone sculpting and surface smoothing. (4) Surgical Drill - Used for bone drilling and screw insertion.

2.2 Advanced Bone Cutting Technologies

Advancements in bone-cutting technology have led to the development of more precise and minimally invasive methods that improve surgical outcomes and reduce risks. Laser-based cutting systems, such as the Er:YAG laser used in the CARLO robotic system, enable highly precise bone ablation while minimizing mechanical stress and heat generation. This is achieved by optimizing pulse duration and laser wavelength, allowing sufficient cooldown time and controlling penetration depth to prevent thermal damage [2]. These systems function by emitting a monochromatic beam of light onto the bone, increasing its temperature and causing ablation through melting or vaporization [2]. Since lasers do not require significant force for manoeuvring, surgeons can achieve higher resection accuracy compared to conventional tools [1]. Additionally, the absence of bone debris prevents obstruction of the surgeon's field of view. By eliminating direct contact with the bone, lasers reduce trauma, post-operative complications, and overall operation time. However, challenges such as high costs and the need for additional cooling mechanisms remain. Furthermore, the presence of water in bone pores can lead to water vaporization, resulting in micro-explosions that weaken bone integrity and create surface defects, potentially compromising implant stability.

Ultrasonic cutting, or piezo surgery, is another emerging technology that utilizes high-frequency vibrations to selectively cut bone while preserving surrounding soft tissues. This technique is particularly advantageous in delicate procedures near nerves and blood vessels, offering improved safety compared to traditional cutting tools.

System	Surgical Application	Autonomy	DOF	Lifespan	Cutting Mechanism	Cooling Requirement	Precision (mm)	Debris Generation
Burr	Spine / Neuro / Orthopedic	Passive	3	50–100 uses	Rotary grinding	High (saline irrigation)	0.5	High
Drill	Orthopedic / Spine	Passive	2–3	100 uses	Rotary cutting	Moderate	0.8	High
Saw	THA / TKA / Cranial	Semi-active	4	50–75 uses	Oscillating cutting	High	1.0	Moderate
Mill	Joint Replacement / Spine	Active	5	75–150 uses	Rotary sculpting	Moderate	0.3	Moderate
Wire Saw	Spine / Pelvic / Craniofacial	Passive	1	30 uses	Abrasive friction	Low	1.5	Low
Laser	CMF / Spinal / Orthopedic	Semi-active	5	2000 hours	Ablation (light energy)	High (cooling system)	0.1	None
Ultrasonic	Spine / Neuro / Orthopedic	Active	4	200 uses	High-frequency vibration	Moderate	0.2	Minimal

Table 1: Comparison of Bone Cutting Systems Based on Surgical Properties

The table outlines essential properties for evaluating the effectiveness of bone-cutting tools in achieving optimal surgical outcomes. Autonomy refers to the level of manual control, ranging from fully manual (passive) to automated (active) systems that enhance precision. Cooling requirements are crucial to prevent thermal necrosis, with high-speed tools needing continuous irrigation or integrated cooling to dissipate heat and protect bone tissue.

3 Task 2: Conformal Tool Design

For this task, we are designing a conformal tool for surgical bone tumor removal. A key requirement is en bloc resection, where the tumor is excised in one piece with a margin of healthy tissue. Fragmentation during removal can spread cancer cells, increasing metastasis risk and complicating recovery. Thus, the tool must prioritize precision and tumor integrity.

We designed an Active Hybrid Ultrasonic-Laser Dual-Axis Milling System combining ultrasonic-assisted milling with laser pre-cutting. The laser defines tumor boundaries with high precision, while ultrasonic milling removes bone with minimal impact. The dual-axis mechanism enables flexible access to complex tumor geometries and positions.

3.1 Dual-Axis Articulating Burr

3.1.1 Dual Axis

Avoiding tool repositioning in surgeries offers benefits like reduced operation time, increased precision, and minimized error risk. In complex surgeries, like en bloc resections, maintaining stable tool position is crucial for accurate margins and healthy tissue preservation. Our solution is a dual-axis articulation mechanism with two movements:

1. **Primary Axis (Depth Control):** Moves the cutting tip up and down, controlling cut depth without overcutting.
2. **Secondary Axis (Angular Control):** Pivots the cutting tip side-to-side, enabling angled or curved cuts for complex tumor geometries.

These movements allow precise multi-plane cuts without repositioning, improving surgical efficiency, accuracy, and safety.

3.1.2 Extender

In deep-seated tumor procedures, the tool's main body can obstruct access to the target area. The extender, integrated into the dual-axis articulating burr, increases the tool's effective length, enabling access to hard-to-reach sites without repositioning.

The extender also provides stability, ensuring consistent control while cutting deeper tissues or bones. Integrated into the dual-axis mechanism, it allows vertical and angular movements at extended lengths, ensuring precise cuts and minimizing disruption to surrounding healthy tissues.

3.2 Surgical Workflow

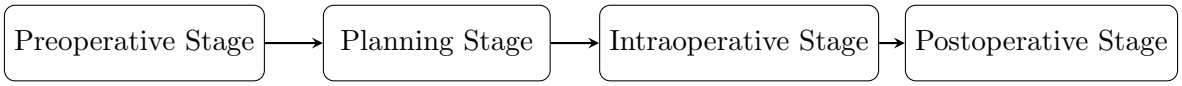


Figure 5: Flowchart of Surgical Workflow

1. Preoperative Stage

High-resolution imaging, such as CT or MRI scans, is utilized to visualize the tumor, surrounding tissues, and anatomical structures. This imaging provides an accurate 3D representation of the tumor's size, shape, and location relative to critical boundaries, such as nerves and blood vessels. Additionally, tumor analysis involves tissue differentiation using QUS and imaging data to distinguish between tumor and healthy tissue. Parameters like scatterer size, acoustic concentration, and echogenicity are analyzed to precisely define the tumor margins.

2. Planning Stage

A safety margin around the tumor is calculated from imaging data to preserve healthy tissue, defining a "critical zone" for guiding the robotic system. Trajectory planning generates an optimal path for the burr, ensuring tumor removal while maintaining safe distances from critical structures. This includes determining the entry point, angle, and precise alignment with the tumor margins. V-shaped cuts are optimized for safety margins, with 3D simulations validating the trajectory. The burr is calibrated with depth settings for vertical cuts, angular adjustments for lateral cuts, and ultrasonic sensors for real-time feedback.

3. Intraoperative Stage

The robotic arm is aligned with the planned trajectory for precise control of the articulating burr. A sterile setup is established, and the burr is securely mounted. During resection, real-time ultrasonic sensor feedback monitors tissue density to assess proximity to tumor margins and healthy structures. Automated safety mechanisms stop the burr if critical boundaries are approached. QUS-based tissue classification, using machine learning algorithms, analyzes data in real time to differentiate tissue types and adjust the cutting path.

The system follows the planned trajectory, making precise depth cuts with the primary axis and angular V-shaped cuts with the secondary axis, ensuring accurate and controlled operation while preserving healthy tissue.

4. Postoperative Stage

After resection, post-operative imaging (CT or MRI) confirms tumor removal and evaluates surrounding tissue preservation. If needed, QUS or other imaging techniques can be used intraoperatively to verify complete excision.

Post-procedure, the patient's recovery is closely monitored to prevent complications from tissue damage or incomplete resection. This stage is crucial for assessing surgical success and ensuring optimal outcomes, with continuous monitoring to address potential issues early.

3.3 CAD Design

Figure 6: Active Hybrid Ultrasonic-Laser Dual-Axis Mill

Figure 7: Ball end milling cutter

3.3.1 Overview of the End-Effector Design

All bone resecting surgical tools share the same drawbacks; mechanical and thermal damage to the bone tissue. Optimizing cutting parameters, such as, the material used for milling, the design of the grooves and the rotational speed of the mill will lead to a more efficient end-effector. Table 2 depicts the thermal conductivity and specific heat capacity of bone and the proposed material for the mill.

Material	**Density** ρ (kg/m^3)	**Specific heat Capacity** $(KJ/(kg*K))$	**Thermal Conductivity k** $(W/(m*K))$
Bone	2030	1.29	0.59
M2 High Speed Steel	78160	470	41

Table 2: Material Thermal Properties [5] [6]

Bone tissue consists of osteon fibres [6], providing strength and flexibility. While bones are structurally anisotropic, they are thermally isotropic, meaning thermal conductivity is uniform. Bones have low specific heat and thermal conductivity, causing heat to accumulate and raising the risk of thermal necrosis during resection. M2 High Speed Steel, with a specific heat capacity of 470 $KJ/(kgK)$ and thermal conductivity of 41 $W/(mK)$, contrasts with bone's properties. This ensures heat generated during cutting is rapidly dissipated, preventing localized overheating and thermal necrosis.

Material	**Ultimate Tensile Strength** (MPa)	**Yield Strength** (MPa)	**Brinell Hardness** (BHN)
M2 High Speed Steel	1200	1000	720
Austentic Stainless Steel	515	205	201

Table 3: Milling material mechanical properties [5] [7]

Regarding the mechanical properties of M2 High Speed Steel, Table 3 depicts the mechanical properties of M2 High Speed Steel and of a typical bone milling tool. M2 High Speed Steel has an ultimate tensile strength (UTS) of more than double than a typical bone milling tool, ensuring the mill can withstand large mechanical loads without fracturing, ensuring durability during high-speed and large cutting force milling. Additionally, having a higher yield strength of 1000 MPa and Brinell Hardness of 720 BHN suggests that the milling tool can maintain its sharp cutting edges under stress, ensuring longevity.

Another important parameter in designing a milling tool for bone resection is the feed rate and the cutting speed. They are both represented by the following equations:

$$f = \frac{v}{N_m n} \quad [2] \tag{3}$$

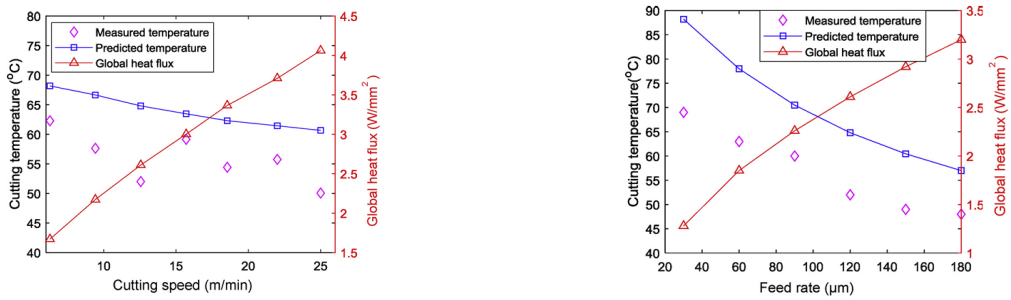
Where f is the feed rate (distance the mill tool moves in one revolution relative to the workpiece), v is the cutting speed of the milling tool, N_m is the rotational speed of the mill and n is the number of teeth on the mill.

$$v = \pi D N_m \quad [2] \tag{4}$$

Where D is the mill diameter.

Due to the large difference in specific heat capacity and thermal conductivity of the two materials, as you increase the cutting speed and the feed rate, the heat transmitted to the bone workpiece reduces [6]. This is because the tool gets into contact with the bone workpiece

more, therefore, more heat from the bone gets transmitted to the mill. To keep the cutting speed and the feed rate high, fewer teeth are required for the mill and the rotational speed of the mill may increase. Figures 8a and 8b exhibit this reduction in temperature.



(a) Against cutting speed (b) Against feed rate

Figure 8: Cutting temperature with respect to a change in cutting speed and feed rate [6]

The milling tool in Fig. 6 & 7 has a cylindrical shape with helical grooves on its surface. A cylindrical mill allows for enhanced manoeuvrability, facilitating precision in challenging areas of the bone. This ensures that the tumour is removed en-bloc. The helical grooves aid in debris (chip) removal as it guides the debris out of the cutting zone in a helical path. This reduces the clogging happening in the cutting zone, thus reducing heat build-up due by lower friction.

Another important physical parameter of the mill is the design of the teeth. For bone resection, multi-tooth corn edge milling is preferred. Multi-tooth corn edge milling reduces the cutting forces by $45.60 - 65.96\%$ compared to straight-edge mills [8], thus decreasing friction and heat build up, reducing the risk of thermal necrosis. Fig. 9 shows the two types of mill designs.

(a) Straight-edge mill (b) Corn-edge mill

Figure 9: Multi-tooth milling [8]

A multi-tooth corn edge mill, as depicted in Fig. 9b, contains a main cutting edge and a secondary cutting edge. The main cutting edge functions like a standard mill, the secondary cutting edge removes burrs (small, sharp fragments of bone left behind after the cutting process) left behind by the main cutting edge (removes a portion of the bone that was left uncut by the previous tooth). The function of having multi-tooth milling also allows many smaller edges to share the cutting load (instead of one large cutting edge engaging with the bone, as in straight-edge traditional milling), reducing the stress on the bone, thus preventing fractures and burrs on and near the cutting zone.
Table 4 exhibits the optimal parameters of the teeth.

Mill type	**No. of teeth**	**Helix angle** (°)	**Peak-to-Peak Tooth Height** (mm)	**Micro Tooth Pitch** (mm)
Straight-edge	3	45	0.00	0.00
Corn-edge	8	18	0.28	2.00

Table 4: Mill type optimal parameters [8]

Micro-tooth pitch is the distance between adjacent micro-teeth peaks, and peak-to-peak tooth height is the vertical distance between the highest and lowest points of the teeth on the cutting edge.

Based on literature, we developed a ball-end milling cutter with two additional degrees of freedom (2-DOF) to handle irregular tumor shapes. A passive tool end may miss certain areas, causing excessive healthy bone removal. The tool uses two micro-servomotors for pitch/yaw control, with harmonic drives for high torque and precision, and an adapter for robotic arm connection. Inverse kinematics calculates the tool head's target angle.

PID ensures angular precision, while a strain gauge provides real-time cutting force feedback. The 2-DOF tool and 6-DOF robotic arm (8-DOF total) support methods like contour cutting, helical cutting, and surface follow-up.

The cutter is made of M2 high-speed steel for thermal conductivity and durability. Its 8-flute corn-cob design improves chip evacuation and cutting efficiency, suitable for hard bone. The cutter is 6.35 mm in diameter and 120 mm in length, ensuring deep reach with minimal vibration.

3.3.2 Operation Workflow of the End-Effector

The ball-end milling cutter, with its spherical geometry and versatile cutting capabilities, is ideal for bone tumor resection. Integrated with an 8-DOF robotic system (2-DOF tool head + 6-DOF robotic arm), it provides precise control and real-time force feedback for minimally invasive removal.

This report analyzes five tumor cases, demonstrating how the cutter, using strategies like contour cutting, helical milling, and layered removal, efficiently handles each scenario. Its ability to adapt to irregular shapes, remove deep tumors, and preserve healthy bone tissue makes it superior to traditional methods.

The following sections detail tumor-specific strategies, showcasing the cutter's precision and efficiency in complex procedures.

Tumor 1: Located near a joint and fully enclosed within the bone. The irregular shape requires precise contour following. With 8 DOF control, the ball-end cutter can easily trace the tumor outline. The procedure involves:

(a) Tumor (dark colour) in the bone (b) Excised section shown in red

Figure 10: Resecting of tumor

1. **Layered cutting** to remove the bone above the tumor and expose the target area.
2. **Contour cutting** for rough removal, which quickly removes most of the tumor.
3. **Helical cutting** for fine detail removal from multiple angles.

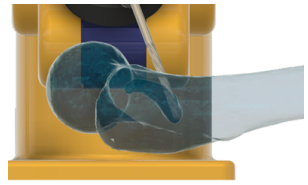
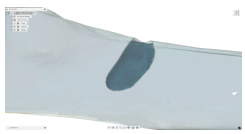


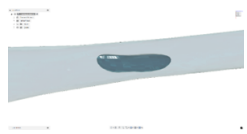
Figure 11: 8-degree-of-freedom milling cutter can cut tumor with different angles

Tumor 2: Cylindrical-shaped tumor.



(a) tumor 2

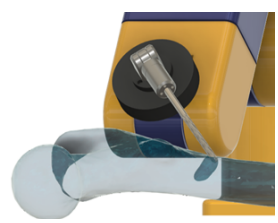
(b) tumor 3



(c) tumor 4

Figure 12: Tumors

(a) helical cutting in different angle



(b) helical cutting in different angle

Figure 13: Helical cutting

The proposed strategy is as follows.

1. Perform **rough cylindrical cutting**, creating a shape that closely matches the tumor profile.
2. Use **helical cutting** to refine the edges and remove residual tissue.

Tumor 3: Elliptical tumor partially exposed on the bone surface. A **helical milling path** is most suitable, starting from the tumor center and gradually expanding outward to remove it entirely.

Tumor 4: A flat, disc-shaped tumor located at a narrow section of the bone. The cutting strategy involves:

1. **Contour cutting** to open the bone and reach the tumor.
2. **Helical cutting** for detail removal.

   Additionally, because the tumor is near a thin section of the bone, the cutting speed should be reduced to prevent fractures.

Tumor 5: Irregular tumor near a joint, similar to Tumor 1. The strategy follows the same approach:

Figure 14: tumor 5

1. Perform **layered contour cutting** to open the bone.
2. Use **helical cutting** for slow and precise removal of the tumor.

3.4 Safety Precautions

Bones subjected to temperatures above $47°C$ for $1; min$ result in thermal necrosis [2]. Therefore, an internal cooling system is employed in the tool, using a saline solution (0.9 $NaCl$ with water) that has higher thermal conductivity than air, efficiently transferring heat away from the bone. The saline solution also lubricates the cutting zone, improving cut smoothness and reducing tool resistance. The open-type internal cooling system releases coolant from the mill tip, directly cooling the cutting zone [2]. The TiO_2-coated internal channel enhances cooling by spreading the coolant evenly, maximizing contact for efficient heat transfer and faster cooling. Fig. 15 shows the effect of TiO_2 coating compared to commercial cooling fluid.

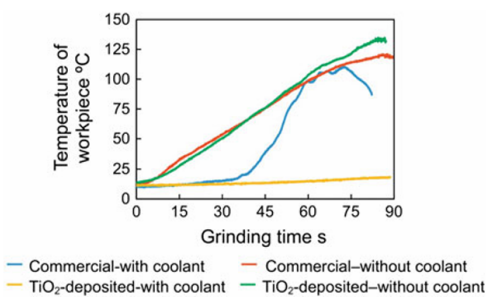


Figure 15: Effect of using TiO_2 coating in cooling system [2]

In Fig. 15 the TiO_2 coating cooling system managed to keep the tool in very low temperatures (below $25°C$) for about $100s$.

Considering after a longer grinding time the temperature will inevitably increase and possibly reach the dangerous maximum of $47°C$, a high-precision infrared sensor near the mill tip monitors real-time bone temperature, processed by a feedback system. This system, controlled by a microcontroller, compares the temperature to the safe threshold. If exceeded, the PID-controlled variable-speed motor reduces the mill's rotational speed, lowering cutting forces to prevent thermal necrosis. As the temperature decreases, the speed gradually increases. Additionally, the coolant flow rate increases to rapidly dissipate heat from the bone.

3.4.1 Ultrasonic Sensor

The integration of an ultrasonic sensor in the articulating burr real-time tissue differentiation. By detecting differences in tissue density, the sensor enables the tool to distinguish

between tumor tissue and healthy bone. This capability ensures that the robotic system effectively targets the tumor while preserving as much healthy tissue as possible. The sensor also provides continuous feedback on the burr's proximity to tumor margins and critical structures, helping the robotic arm maintain precise control over the resection process.

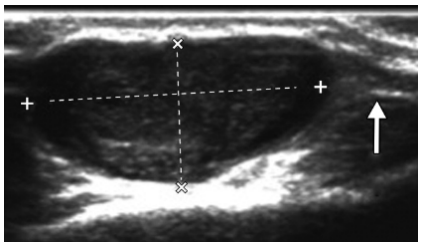


Figure 16: Ultrasonic imaging demonstrating tissue differentiation by detecting echogenicity variations and identifying tumor margins [9]

The ultrasonic sensor, positioned behind the cutting tip, acts as a critical safety mechanism by monitoring the distance between the burr and surrounding tissues. If it detects proximity to healthy tissue or critical boundaries, it triggers an automatic halt to prevent damage, ensuring a controlled procedure. Integrated at the burr's base, the sensor provides real-time feedback without interfering with cutting.

To enhance functionality, Quantitative Ultrasound (QUS) and machine learning can be used for detailed tissue analysis. QUS analyzes tissue microstructures through parameters like scatterer diameter and acoustic concentration, enabling precise differentiation between healthy and pathological tissues, especially for tumor margin identification.

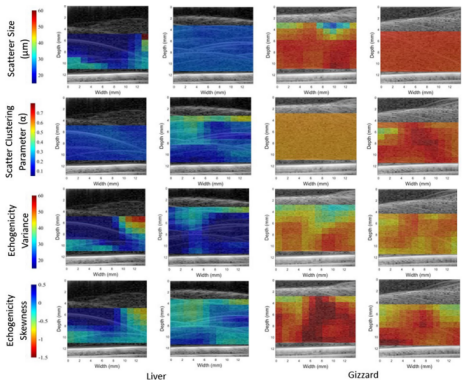


Figure 17: QUS parameter overlays for liver and gizzard samples, showing scatterer size, clustering, echogenicity variance, and skewness, with color-coded maps over the B-mode ultrasound image. [10]

The QUS parameter overlays identify tissue characteristics, distinguishing healthy and pathological regions, as shown in Fig. 17. This data helps define a "critical zone" around the tumor, ensuring the robotic system operates within safe boundaries. Integrating these overlays allows the articulating burr to avoid healthy tissue, maintaining precise resection margins and preserving critical structures.

With machine learning algorithms like K-nearest neighbors (KNN), QUS data can be processed for near-real-time tissue classification with high accuracy. Studies have shown 94.5

4 Task 3: Surgical Planning

For the double axis burr, the cutting pattern designed is circular milling layer by layer. This is better than block milling. Instead of cutting a large rectangular block, which could damage healthy bone, circular milling allows for gradual material removal. It follows a layered approach, reducing cutting forces and heat buildup. A conical helix drill is also well-suited for circular cuts. It allows for continuous tool engagement, leading to less vibration and better surface finish. Layer-by-layer cutting can conform better to the tumor shape rather than using an arbitrary rectangular box. This ensures minimal bone removal while still extracting the full tumor. It further makes automation easier because it avoids the sharp corners that may cause tool deflection in block cutting.

Instead of computing a bounding box, the program now computes concentric circular layers around the tumor region.

4.1 Parameterize the Cuts

To parameterize the cuts, we need to define:

- Position (x, y, z): Where to start and end the cut.
- Orientation (θ, ϕ): The angles of the tool relative to the bone surface.
- Cut Shape & Size: Circular milling path that conforms to the tumor.
- Cut Depth: Final depth of the tool.
- Layer Thickness (Δz): The vertical step for layer-by-layer cutting.

The first step is to extract the tumor geometry from the imaging data using segmentation software. This should indicate the boundaries of the tumor and sections embedded in bone, showing the best places to start and end milling.

4.1.1 Position (Entry Point Calculation)

4.1.2 Principal Component Analysis (PCA) for Tumor Orientation

The purpose of PCA is to find the principal axes of the tumor so that we can align the cutting tool optimally. The covariance matrix captures how the points are distributed in 3D space. By centering the points at the centroid (mean of the points), the covariance matrix is calculated as:

$$\Sigma = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{T}_i - \bar{\mathbf{T}})(\mathbf{T}_i - \bar{\mathbf{T}})^T$$

where:

- $\mathbf{T}_i$ represents the individual tumor points (in 3D).
- $\bar{\mathbf{T}}$ is the centroid of the tumor (calculated as the mean of all tumor points).

From the covariance matrix, the eigenvectors represent the principal directions of the data, and the eigenvalues represent the magnitude of variance along those directions. The largest eigenvector corresponds to the major axis (the longest axis), and the smallest one corresponds to the minor axis (the shortest axis).

For the tumor's alignment:

- V_1 is the longest axis (major axis).
- V_2 is the second-longest axis.
- V_3 is the shortest axis.

The axis with the largest eigenvalue (λ_1) is considered the dominant principal direction of the tumor. Once PCA is done and the orientation is known, the centroid of the tumor is calculated as the average of all tumor points (X, Y, Z). The centroid serves as a reference for defining milling paths. This is calculated as:

$$C_x = \frac{1}{N}\sum_{i=1}^{N} X_i, \quad C_y = \frac{1}{N}\sum_{i=1}^{N} Y_i, \quad C_z = \frac{1}{N}\sum_{i=1}^{N} Z_i$$

Where:

- N is the total number of tumor points.

4.1.3 Determining the Radius at the Cut Layer

For each layer of milling, the maximum distance from the centroid to points at that specific Z-height cut_z gives us the radius for the milling path. For each point on the tumor at that height, we calculate:

$$r_{\text{tumor}} = \max\left(\sqrt{(X_i - C_x)^2 + (Y_i - C_y)^2} \mid Z_i = \text{cut}_z\right)$$

This calculates the maximum radial distance at the current Z-height.

4.1.4 Adding a Safe Offset

A small offset is added to the computed radius to ensure that the milling tool stays outside the tumor and doesn't cut into it. This offset is a predefined constant (e.g., 3 units):

$$r_{\text{cut}} = r_{\text{tumor}} + \text{offset}$$

The value of r_{cut} is the actual milling radius for that layer.

4.1.5 Calculating the Entry Point

The milling tool entry point is located at the distance r_{cut} from the centroid at $z = 0$ (the surface of the bone). This ensures that the milling starts from the correct distance from the tumor's center. The entry point is represented as:

$$P_{\text{entry}} = (C_x + r_{\text{cut}} \cdot \cos(\theta), C_y + r_{\text{cut}} \cdot \sin(\theta), 0)$$

Where:

- θ is the angle for the circular milling path (ranging from 0 to 2π).
- r_{cut} is the radius calculated previously.

This entry point lies on the circular milling path at the surface of the bone, which is at $z = 0$.

4.1.6 Determining the Final Cut Point

The deepest point of the tumor inside the bone determines how far we mill:

$$z_{\text{finish}} = \min(z_{\text{tumor}})$$

where z_{tumor} are the tumor's unique vertices.

4.2 The Orientation of the Tool

The orientation of the tool is defined by two angles:

- Tilt Angle (θ): Angle between the tool axis and the bone surface normal.
- Rotation Angle (ϕ): Rotation of the tool around the vertical axis.

4.2.1 Tilt Angle (θ)

At each milling layer, the tool should be aligned with the tumor's longest axis and perpendicular to the bone:

- The tumor's longest axis is given by the first principal component:

$$\mathbf{V}_1 = (V_{1x}, V_{1y}, V_{1z})$$

- The bone surface normal at $z = 0$ is $(0, 0, 1)$ (along the Z-axis).
- The tilt angle θ is the angle between $\mathbf{V}_1$ and the bone normal:

$$\theta = \cos^{-1}(V_{1z})$$

  where V_{1z} is the Z-component of the longest axis. If $V_{1z} = 0$, the tool remains perfectly vertical.

4.2.2 Rotation Angle (ϕ)

The rotation angle ϕ aligns the tool with the longest axis in the XY-plane:

$$\phi = \tan^{-1}\left(\frac{V_{1y}}{V_{1x}}\right)$$

where (V_{1x}, V_{1y}) define the direction of the longest axis in the XY-plane. This ensures the tool is oriented correctly in the milling plane.

Since the tumor might be slanted inside the bone, the orientation gradually adjusts as the tool moves deeper. At each milling depth z_k:

- Compute the local normal of the bone at z_k (if available).
- Adjust θ based on how the tumor extends at deeper layers.
- Maintain ϕ along the major axis unless bone curvature requires adjustments.

If the bone surface is assumed flat ($z = 0$ is constant), then θ remains constant. Otherwise, for curved bones, local normals should be used.

4.3 Determine the Cutting Depth at Each Layer

The milling process goes from the bone surface ($z = 0$) down to the deepest tumor point. The depth levels are defined as:

$$z_k = z_0 - k \cdot \Delta z$$

where:

- Δz is the layer thickness (step-down distance per pass).
- k is the current layer index – the layer of the milling process currently at.
- The process stops at $z_{\min}$ (deepest tumor point):

$$z_k \geq z_{\min}$$

Key Considerations:

- Smaller $\Delta z \rightarrow$ More precision, but slower milling.
- Larger $\Delta z \rightarrow$ Faster, but may cause excessive bone removal.

4.4 Circular Path Planning

Since we are doing layer-by-layer milling, each layer follows a circular milling path around the tumor, ensuring a safe offset. The circular path at each depth z_k is defined as:

$$x_k = x_{\text{milling},k} + R_{\text{milling},k} \cos\theta$$

$$y_k = y_{\text{milling},k} + R_{\text{milling},k} \sin\theta$$

$$z_k = z_k$$

where:

- $(x_{\text{milling},k}, y_{\text{milling},k})$ is the milling center at depth z_k (this could be the tumor centroid at that layer).
- $R_{\text{milling},k}$ is the radius of the milling path at that layer.
- $\theta \in [0, 2\pi]$ defines a full circular path.
- z_k is the current milling depth.

Additionally, parameters such as cutting speed and feed rate can be determined on a case-by-case basis.

4.5 Rectangular Blocks for Tumor Removal

Circular cuts are well-suited for irregular or rounded tumor shapes because they provide flexibility in conforming to complex geometries. However, in cases where a rectangular bounding box is necessary to enclose the tumor, the following parameterization steps are used to plan the cuts:

4.6 Position: Entry and Exit Points

- The entry point is calculated as before, ensuring the tool starts at a position on the bone surface ($z = 0$), with a slight offset to avoid direct contact with the tumor.
- The exit point is similarly determined, considering the tool's path and ensuring it completes the cut at the appropriate depth, leaving a clean and precise boundary.
- The tool path between these points is calculated to ensure smooth engagement with the bone, avoiding any contact with the tumor itself.

4.7 Orientation: Tool Alignment for Cutting

- **Vertical Cuts:** The vertical cuts are made along the longest axis of the rectangular block, which is aligned perpendicular to the bone surface. This ensures that the tumor is isolated from the surrounding bone with minimal disruption.
- **Horizontal Cuts:** The horizontal cuts, which define the bottom of the tumor block, are performed using the tool's double-axis capability. The tool moves parallel to the bone surface and cuts just below the tumor's minimum depth, ensuring that the tumor is detached from the bone efficiently. This horizontal cut removes bone material without affecting the tumor itself.

4.8 Width of the Block

The width of the rectangular block is determined by the distance between the tumor's leftmost and rightmost points in the x-axis direction. This ensures the block fully encompasses the tumor without excess bone removal on the sides:

$$\text{Width} = x_{\max} - x_{\min}$$

where:

- $x_{\min}$ and $x_{\max}$ are the minimum and maximum x-coordinates of the tumor boundary.

4.9 Length of the Block

The length of the rectangular block is defined by the distance between the tumor's front and rear points in the y-axis direction.

$$\text{Length} = y_{\max} - y_{\min}$$

where:

- $y_{\min}$ and $y_{\max}$ are the minimum and maximum y-coordinates of the tumor boundary.

4.10 Breadth of the Block

The breadth of the rectangular block corresponds to the depth of the tumor along the z-axis, extending from the surface of the bone down to the deepest point of the tumor.

$$\text{Breadth} = z_{\max} - z_{\min}$$

where:

- $z_{\min}$ and $z_{\max}$ are the minimum and maximum z-coordinates of the tumor boundary, ensuring that the block fully encloses the tumor's depth.

5 Pseudocode for Parametrization of the Cutting Path

```
1. Load Tumor Data
    - Load tumor point cloud data (X, Y, Z coordinates of tumor
       points).

2. Principal Component Analysis (PCA) for Orientation
    - Compute the centroid of the tumor:
        Cx = (1/N) * ∑X_i
        Cy = (1/N) * ∑Y_i
        Cz = (1/N) * ∑Z_i
        Where N is the number of tumor points.

    - Compute the covariance matrix of the tumor points:
        Σ = (1/N) * ∑[(T_i − C)(T_i − C)^T]

    - Perform PCA on the covariance matrix:
        - Compute the eigenvectors (V1, V2, V3) and eigenvalues (λ_1
           , λ_2, λ_3) from the covariance matrix.
        - Sort eigenvalues such that λ_1 ≥ λ_2 ≥ λ_3.
        - Assign V1 to the longest axis (major axis), V2 to the
           second-longest axis, and V3 to the shortest axis.
```

```
    - Orientation Determination:
        - The longest axis (V1) corresponds to the cutting
          orientation.
        - The direction of V1 (e.g., if the longest axis is along
          the Z-axis) will define the milling orientation.
        - Based on the orientation of V1, adjust the milling path
          accordingly (rotate the milling path if necessary to
          align with V1).

3. Define Milling Layers
    - Define the number of milling layers num_layers.
    - Create layer heights, dividing the range from the top of the
      bone (z=0) to the bottom of the tumor.
    - Use linspace() to generate evenly spaced layer heights.

4. Iterate Through Layers (For each layer)
    For each layer height cut_z:

    a. Find Points at Current Layer
        - Identify all tumor points at cut_z (within a small range
          of ±0.5).

    b. Calculate Maximum Radius at Current Layer
        - For each tumor point at cut_z, compute the distance from
          the centroid in the XY-plane:
            distance = √((X_i − C_x)^2 + (Y_i − C_y)^2)
        - Find the maximum radius r_tumor for the current layer:
            r_tumor = max(distance)

    c. Add Safety Margin
        - Add a small safety margin (radius_offset) to ensure the
          milling tool stays outside the tumor:
            r_cut = r_tumor + offset

    d. Generate Circular Milling Path
        - Calculate the circular milling path for the current layer
            at cut_z:
            x_cut = C_x + r_cut * cos(θ)
            y_cut = C_y + r_cut * sin(θ)
            Where θ ranges from 0 to 2π (a full circle).
        - Plot the milling path in the XY-plane at cut_z.

    e. Adjust Orientation of Milling Path (Align with PCA
       orientation)
        - If V1 (the major axis) is not aligned with the Z-axis,
          rotate the milling path by the PCA rotation matrix:
            - Rotate the milling circle coordinates (x_cut, y_cut) by
              the rotation matrix that aligns V1 with the Z-axis.

5. Final Horizontal Pass
    - Deep Milling Pass:
        - After milling through layers, generate a horizontal cut
          plane at a deeper depth (e.g., z_finish = min(Z) − 1.5).
        - The plane extends outward from the centroid in the XY-
          plane based on the maximum radius r_cut and defines the
```

```
            final undercut.

6. Visualize the Milling Process
    - Plot the tumor, milling paths, and the final horizontal plane
      .
    - Ensure all steps (cutting and final plane) are shown in the 3
      D plot.

7. End of Milling Process
    - Complete the milling process and visualize the final shape
      after safe removal.
```

6 Algorithm Implementation on Given Dataset

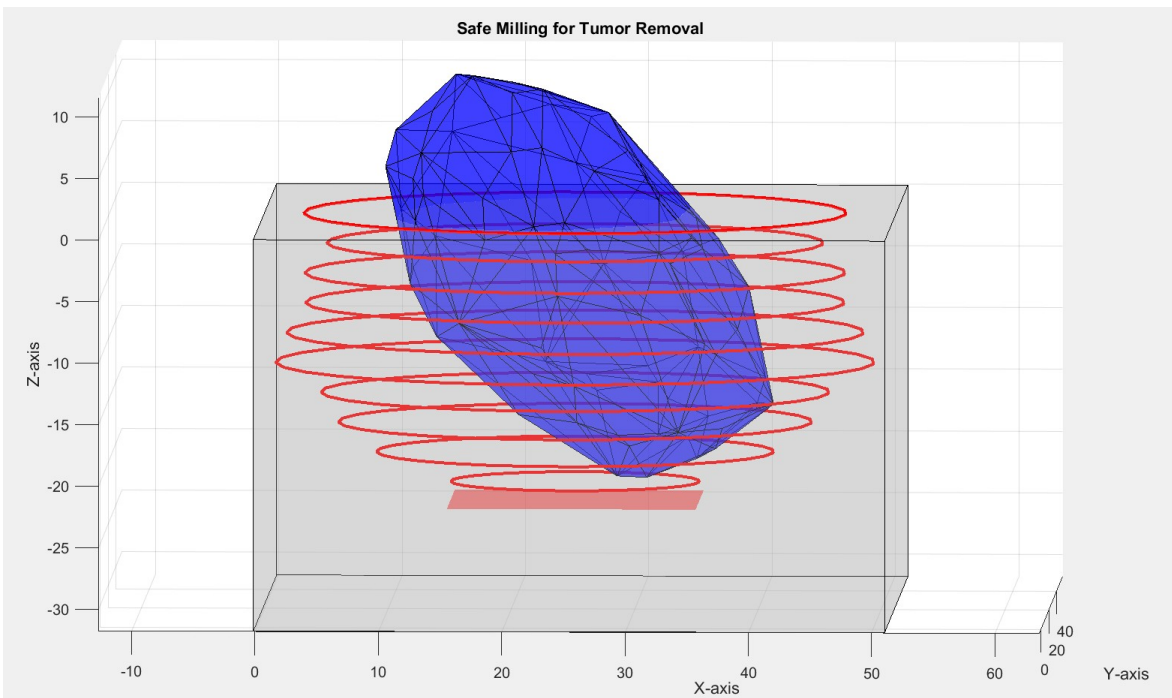


Figure 18: Visualization of the tumor and milling path

The algorithm generates a 3D model based on the provided dataset. The gray cubic enclosure represents the portion of the tumor surrounded by bone. The red circular traces depict the tool's milling path at each layer. At the bottom, the red horizontal plane marks the final cut that will detach the tumor from the bone, allowing for its removal.

7 Conclusion

This report has systematically explored the development of a precise and efficient approach for bone tumor resection, integrating literature research, tool design, and surgical planning.

In Task 1, we analyzed conventional bone-cutting tools, including saws, drills, burrs, and milling devices, discussing their effectiveness and limitations. We also explored advanced cutting technologies such as laser and ultrasonic systems, which offer improved precision, reduced heat generation, and minimized mechanical trauma to healthy tissues.

Task 2 focused on developing an innovative Dual-Axis Articulating Burr, which enhances maneuverability and precision in surgical applications. The tool design incorporated both depth and angular control, ensuring accurate resection while minimizing damage to surrounding bone structures. The design process included CAD modeling, safety mechanisms such as ultrasonic sensors, and real-time feedback systems to enhance cutting accuracy and prevent unintended tissue damage.

In Task 3, we detailed the methodology for parametrizing the cutting path, using PCA to determine the tumor's orientation and define optimal milling strategies. A layer-by-layer circular milling approach was selected to ensure controlled and efficient tumor removal while preserving as much healthy bone as possible. The cutting parameters, including entry and exit points, cutting depth, and tool alignment, were carefully calculated to optimize surgical outcomes.

The Pseudocode and Algorithm Implementation section provided a structured computational framework for executing the surgical procedure. Through the application of the developed algorithm, we successfully generated a 3D visualization of the tumor and milling paths, validating the effectiveness of our approach.

Overall, this report presents a comprehensive strategy for robotic-assisted bone tumor resection, combining innovative tool design, precise surgical planning, and algorithmic execution. The proposed methodology aims to improve surgical precision, enhance patient safety, and optimize clinical outcomes. Future research could focus on further refining the robotic control system, improving real-time feedback integration, and validating the approach through experimental trials on simulated or real surgical scenarios.

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